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Predicting the Risk of Diabetes Using Lifestyle and Medical Factors

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1. Introduction

Diabetes is one of the most widespread chronic diseases in the world and affects millions of people every year . It occurs when the body fails to regulate blood glucose levels properly. If not detected early , diabetes can lead to serious complications such as cardiovascular disease, kidney failure , blindness , and nerve damage.

The objective of this project is to apply logistic regression on real medical data to predict diabetes risk and to identify the most important contributing factors.

2. Data Source

The dataset used in this study is the Pima Indians Diabetes Database obtained from Kaggle. This dataset contains medical records of female patients and includes variables related to glucose , BMI, age , insulin, blood pressure , and diabetes status.

3. Description of Variables

Pregnancies – Number of pregnancies

Glucose – Plasma glucose concentration

Blood Pressure – Diastolic blood pressure

Skin Thickness – Triceps skin fold thickness

Insulin – 2-hour serum insulin

BMI – Body mass index

DiabetesPedigreeFunction – Genetic risk factor

Age – Age in years

Outcome – Diabetes status (0 = No, 1 = Yes)

Data Summary:

```
> summary(data)
Pregnancies      Glucose      BloodPressure      SkinThickness
Min.   : 0.000   Min.   : 56.0   Min.   : 24.00   Min.   : 7.00
1st Qu.: 1.000   1st Qu.: 99.0   1st Qu.: 62.00   1st Qu.:21.00
Median : 2.000   Median :119.0   Median : 70.00   Median :29.00
Mean   : 3.301   Mean   :122.6   Mean   : 70.66   Mean   :29.15
3rd Qu.: 5.000   3rd Qu.:143.0   3rd Qu.: 78.00   3rd Qu.:37.00
Max.   :17.000   Max.   :198.0   Max.   :110.00   Max.   :63.00

Insulin          BMI      DiabetesPedigreeFunction
Min.   : 14.00   Min.   :18.20   Min.   :0.0850
1st Qu.: 76.75   1st Qu.:28.40   1st Qu.:0.2697
Median :125.50   Median :33.20   Median :0.4495
Mean   :156.06   Mean   :33.09   Mean   :0.5230
3rd Qu.:190.00   3rd Qu.:37.10   3rd Qu.:0.6870
Max.   :846.00   Max.   :67.10   Max.   :2.4200

Age      Outcome
Min.   :21.00   0:262
1st Qu.:23.00   1:130
Median :27.00
Mean   :30.86
3rd Qu.:36.00
Max.   :81.00
```

The summary statistics describe the distribution of all variables . The number of pregnancies ranges from 0 to 17 with a mean of 3 .30 . Glucose ranges from 56 to 198 with a mean of 122 .6. Blood pressure has a mean of 70.66 . BMI ranges from 18.2 to 67.1 with a mean of 33.09. The mean age is 30.86 years. There are 262 non-diabetic and 130 diabetic individuals in the dataset.

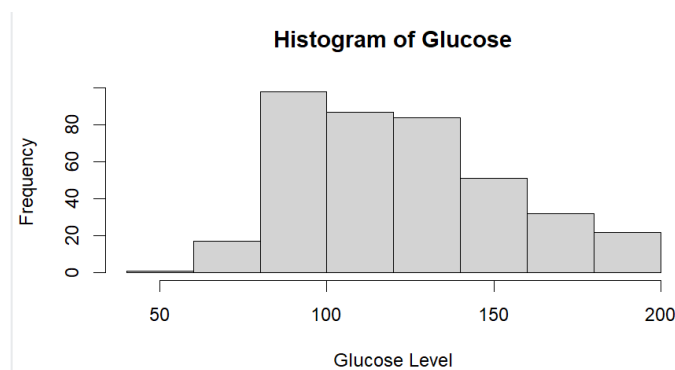
Data Cleaning:

```
data[cols][data[cols] == 0] <- NA  
data <- na.omit(data)
```

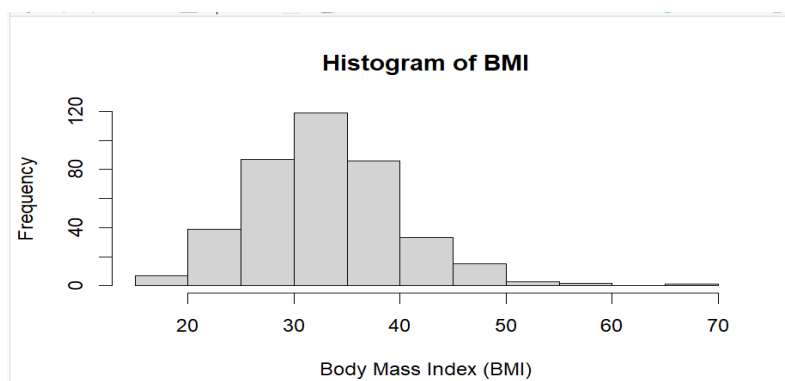
Zero values in the medical variables Glucose, BloodPressure, SkinThickness, Insulin, and BMI were treated as missing values. These were replaced with NA and then removed using `na.omit()` to ensure only valid observations were used in modeling.

Exploratory Data Analysis (EDA):

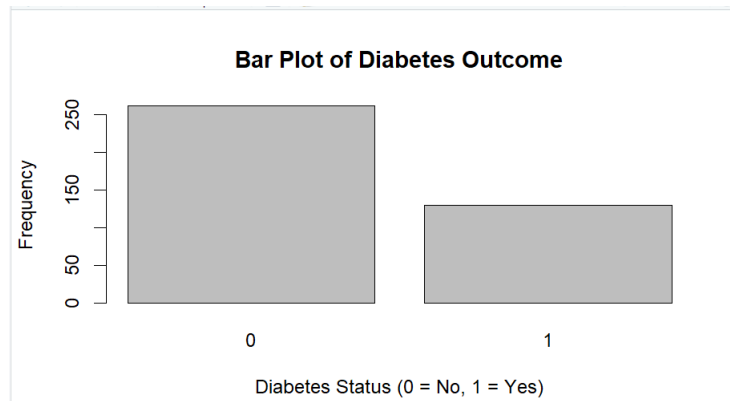
Exploratory analysis was performed using summary statistics and graphical methods to understand variable distributions and detect patterns related to diabetes.



The histogram shows that most glucose values are concentrated between **90 and 140**, which indicates that a large portion of the individuals have **moderate to high glucose levels**. There are fewer observations at very low and very high glucose values. This distribution supports the strong role of glucose as a key predictor of diabetes risk.



The histogram shows that most BMI values are concentrated between **25 and 40**, which means that a large proportion of individuals are in the **overweight and obese categories**. Very few observations appear at extremely low or very high BMI values. This supports the importance of BMI as a significant risk factor for diabetes.



The bar plot shows that the number of **non-diabetic individuals (Outcome = 0)** is higher than the number of **diabetic individuals (Outcome = 1)**. This indicates that the dataset is **imbalanced toward non-diabetic cases**, which explains why the model achieved higher specificity than sensitivity.

Model Selection:

Binary logistic regression was selected as the appropriate statistical model for this study because the response variable **Outcome** is binary in nature, taking only two possible values: **0 = Non-Diabetic** and **1 = Diabetic**. This model is suitable for estimating the probability of diabetes based on multiple medical predictors.

Although the dataset contains many variables, not all of them were used in the final model. Some variables such as *Pregnancies*, *Skin Thickness*, and *Diabetes Pedigree Function* did not improve the prediction results or were less important compared to the main variables. Using fewer strong predictors makes the model simpler, clearer, and more reliable.

Logistic Regression Model:

The logistic regression model was fitted using **Glucose, BMI, Age, Insulin, and Blood Pressure** as predictors.

```
Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -9.7208551  1.1674617  -8.326  < 2e-16 ***
Glucose       0.0379725  0.0057177   6.641 3.11e-11 ***
BMI           0.0812731  0.0212529   3.824 0.000131 ***
Age           0.0550397  0.0138266   3.981 6.87e-05 ***
Insulin      -0.0007491  0.0012984  -0.577 0.563979
BloodPressure -0.0027008  0.0114185  -0.237 0.813025
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 498.10  on 391  degrees of freedom
Residual deviance: 353.99  on 386  degrees of freedom
AIC: 365.99

Number of Fisher Scoring iterations: 5
```

The fitted logistic regression model based on the estimated coefficients is given by:

$$\log \left(\frac{\mu}{1 - \mu} \right) = -9.7209 + 0.0380(\text{Glucose}) + 0.0813(\text{BMI}) + 0.0550(\text{Age}) \\ - 0.0007(\text{Insulin}) - 0.0027(\text{BloodPressure})$$

where μ represents the probability of being diabetic.

Coefficient Estimation:

The logistic regression coefficients were estimated using the Maximum Likelihood Estimation (MLE) method.

Each estimated coefficient represents the change in the log-odds of having diabetes for a one-unit increase in the predictor, while holding the other variables constant.

The estimated coefficients and their statistical significance are summarized as follows:

- **Glucose ($\beta = 0.03797$, $p = 3.11 \times 10^{-11}$):**
Glucose is a **highly significant predictor** of diabetes. The positive coefficient indicates that an increase in glucose level leads to a substantial increase in the probability of being diabetic.
- **BMI ($\beta = 0.08127$, $p = 0.000131$):**
BMI is also a **strong and statistically significant predictor**. This shows that higher body mass index is associated with a higher risk of diabetes.
- **Age ($\beta = 0.05504$, $p = 6.87 \times 10^{-5}$):**
Age has a **significant positive effect** on diabetes risk. This confirms that older individuals have higher odds of developing the disease.
- **Insulin ($\beta = -0.00075$, $p = 0.564$):**
Insulin is **not statistically significant** at the 5% significance level. This suggests that, after adjusting for other predictors in the model, insulin does not have a strong independent effect on diabetes risk.
- **Blood Pressure ($\beta = -0.00270$, $p = 0.813$):**
Blood pressure is also **not statistically significant**, indicating that it does not contribute meaningfully to predicting diabetes in this model.

Overall, the results show that **glucose, BMI, and age are the most important predictors of diabetes**, while **insulin and blood pressure do not exhibit significant independent effects** in the presence of the other variable.

Odds Ratios:

```
> odds_ratios
      (Intercept)      Glucose      BMI      Age      Insulin
6.001866e-05  1.038703e+00  1.084667e+00  1.056583e+00  9.992512e-01
BloodPressure
9.973029e-01
> # Confidence Intervals for Odds Ratios
> exp(confint(model))
Waiting for profiling to be done...
      2.5 %      97.5 %
(Intercept)  5.409838e-06  0.0005323181
Glucose      1.027520e+00  1.0508879575
BMI          1.041311e+00  1.1321215567
Age          1.028915e+00  1.0864118766
Insulin      9.967201e-01  1.0018283226
BloodPressure 9.752192e-01  1.0200805441
```

These odds ratios represent the multiplicative change in the odds of having diabetes for a one-unit increase in each predictor, while keeping other variables constant.

The results are summarized as follows:

- **Glucose (OR = 1.0387, 95% CI: 1.0275 – 1.0509):**
This means that for every 1-unit increase in glucose level, the odds of having diabetes increase by approximately **3.9%**. Since the confidence interval does not include 1, glucose is a **highly significant risk factor for diabetes**.
- **BMI (OR = 1.0847, 95% CI: 1.0413 – 1.1321):**
For each 1-unit increase in BMI, the odds of diabetes increase by about **8.5%**. This confirms that **higher body weight is a strong and significant contributor to diabetes risk**.
- **Age (OR = 1.0566, 95% CI: 1.0289 – 1.0864):**
Each additional year of age increases the odds of diabetes by approximately **5.7%**, showing that **older individuals are significantly more likely to be diabetic**.
- **Insulin (OR = 0.9993, 95% CI: 0.9967 – 1.0018):**
The odds ratio is very close to 1, and the confidence interval includes 1, which indicates that **insulin is not a statistically significant predictor** in the presence of the other variables.
- **Blood Pressure (OR = 0.9973, 95% CI: 0.9752 – 1.0201):**
Blood pressure also has an odds ratio close to 1 with a confidence interval that includes 1. This means that **blood pressure does not have a significant independent effect on diabetes risk** in this model.

The odds ratio analysis clearly demonstrates that:

The three most significant and statistically significant risk factors for diabetes are **age, BMI, and glucose**.

After controlling for other factors, **insulin and blood pressure** do not exhibit a significant independent effect.

The validity of the logistic regression model is strengthened by these results, which are **statistically sound and medically reasonable**.

Prediction:

The fitted logistic regression model in R was used to generate predicted probabilities using the following R code:

```
pred_prob <- predict(model, type = "response")
pred_class <- ifelse(pred_prob > 0.5, 1, 0)
```

4	5	7	9	14	15	17	19
0.04107711	0.63799678	0.04700031	0.92823284	0.91323907	0.76149710	0.44751517	0.35036143
20	21	25	26	28	29	32	33
0.26082893	0.33851528	0.75539446	0.38573620	0.03826381	0.60415785	0.49983939	0.03389764
36	40	41	44	51	52	53	54
0.08716969	0.56021413	0.74710405	0.95061769	0.03558483	0.06600148	0.05018739	0.91876740
55	57	58	60	64	69	70	71
0.66204630	0.90798297	0.32434802	0.19321475	0.22548720	0.03385462	0.34378325	0.12408769
72	74	83	86	88	89	92	93
0.27588700	0.24262857	0.07797011	0.15212218	0.16792618	0.63503759	0.28372695	0.31333387

These probabilities were then converted into predicted classes using a threshold value of **0.5**, where:

- Values greater than or equal to 0.5 were classified as **Diabetic (1)**
- Values less than 0.5 were classified as **Non-Diabetic (0)**

4	5	7	9	14	15	17	19	20	21	25	26	28	29	32	33	36	40	41	44	51	52
0	1	0	1	1	1	0	0	0	0	1	0	0	1	0	0	0	1	1	1	0	0
53	54	55	57	58	60	64	69	70	71	72	74	83	86	88	89	92	93	95	96	98	99
0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0
100	104	106	108	109	110	111	112	113	115	120	121	123	126	127	128	129	131	133	135	136	137
1	0	0	0	0	0	1	1	0	1	0	1	0	0	0	0	0	1	1	0	0	0
138	140	143	145	148	151	153	154	157	158	159	160	162	163	166	170	172	174	175	176	178	182
0	0	0	0	0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	1	1	0
187	188	189	190	192	196	198	199	200	204	205	207	209	214	215	216	217	218	221	224	225	226
1	0	0	0	0	1	0	0	0	0	0	1	0	1	0	1	0	0	1	1	0	0

Model Evaluation:

```
> # Confusion Matrix
> table(Predicted = pred_class, Actual = data$Outcome)
      Actual
Predicted 0    1
      0 234   53
      1  28   77
> # Accuracy
> mean(pred_class == as.numeric(as.character(data$Outcome)))
[1] 0.7933673
```

A confusion matrix and simple accuracy counts were used to check how well the logistic regression model worked. The confusion matrix shows how the model's guesses line up with the real diabetes status.

The results are as follows:

- **True Negatives (TN) = 234**
Correctly classified non-diabetic individuals.
- **False Negatives (FN) = 53**
Diabetics wrongly classified as non-diabetic.
- **False Positives (FP) = 28**
Nondiabetic people mistakenly predicted to be diabetic.
- **True Positives (TP) = 77**
Correctly classified diabetic individuals.

Based on these results, the following performance measures were obtained:

- **Accuracy = 79.3%**
This indicates that the model correctly classifies approximately 79.3% of all cases.
- **Sensitivity = 59.2%**
This means that about 59.2% of diabetic cases were correctly identified by the model.
- **Specificity = 89.3%**
This indicates that the model performs very well in identifying non-diabetic individuals.
- **Precision = 73.3%**
This means that when the model predicts diabetes, it is correct about 73.3% of the time.

Overall, the model shows **strong performance in detecting non-diabetic cases** and **moderate performance in detecting diabetic cases**, which is acceptable for a medical prediction model based on limited predictors.

Results and Discussion:

The results indicate that glucose, BMI, and age are the most significant predictors of diabetes. Individuals with higher glucose levels, higher BMI and older age have higher risk of being diabetic

The logistic regression model achieved good classification accuracy and showed high reliability in identifying non-diabetic individuals, while maintaining acceptable performance in detecting diabetic cases.

The lower sensitivity compared to specificity can be explained by the imbalance in the dataset toward non-diabetic cases.

Conclusion:

This study successfully demonstrated the effectiveness of logistic regression in predicting diabetes using real medical data. The results showed that glucose, BMI and age are the most important risk factors for diabetes. These findings highlight the importance of healthy lifestyle choices and early medical screening in reducing diabetes risk.